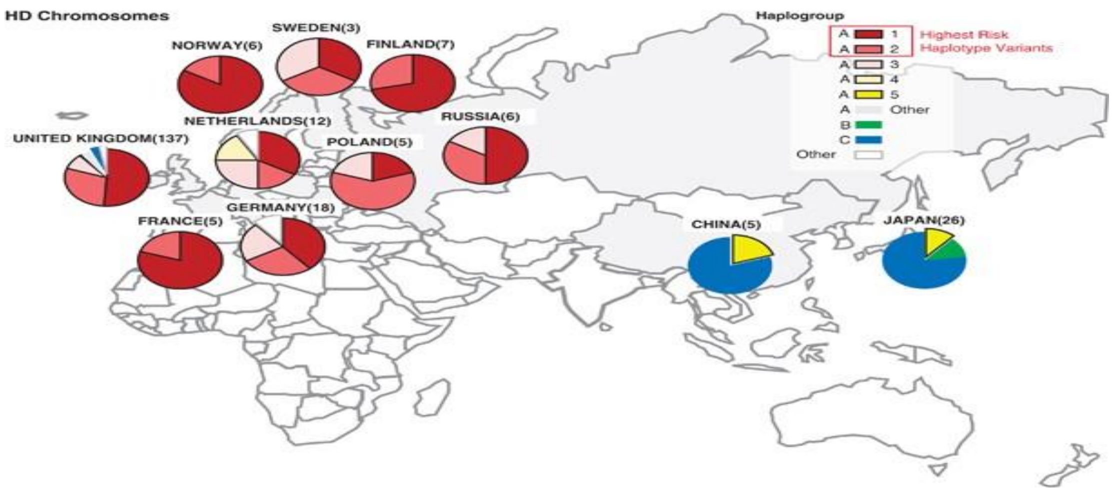
HUNTINGTON'SDISEASE

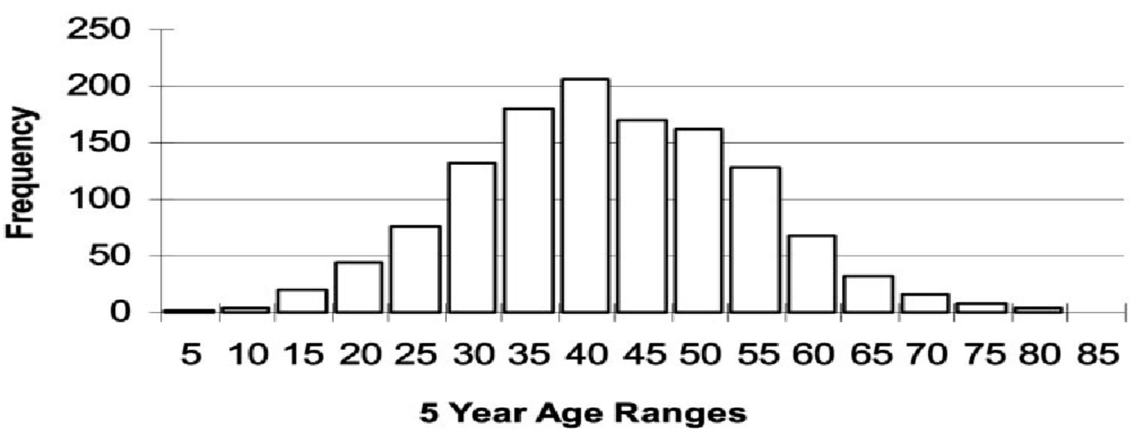
1. Definition

Huntington's disease is a progressive neurodegenerative disorder characterized by involuntary movements (**chorea**), psychiatric symptoms, and cognitive impairment. It results from a mutation in the **HTT gene**, causing production of an abnormal huntingtin protein that gradually damages neurons.

2. Epidemiology

- Rare disorder
- Prevalence: approximately **5–10 per 100,000** people in populations of European ancestry
- Equal frequency in males and females
- Typical onset: **30–50 years**
- Juvenile Huntington's disease: onset before age 20 (less common)





tington's disease onset ages. The age at onset distribution in Huntington's disease is very broad and may

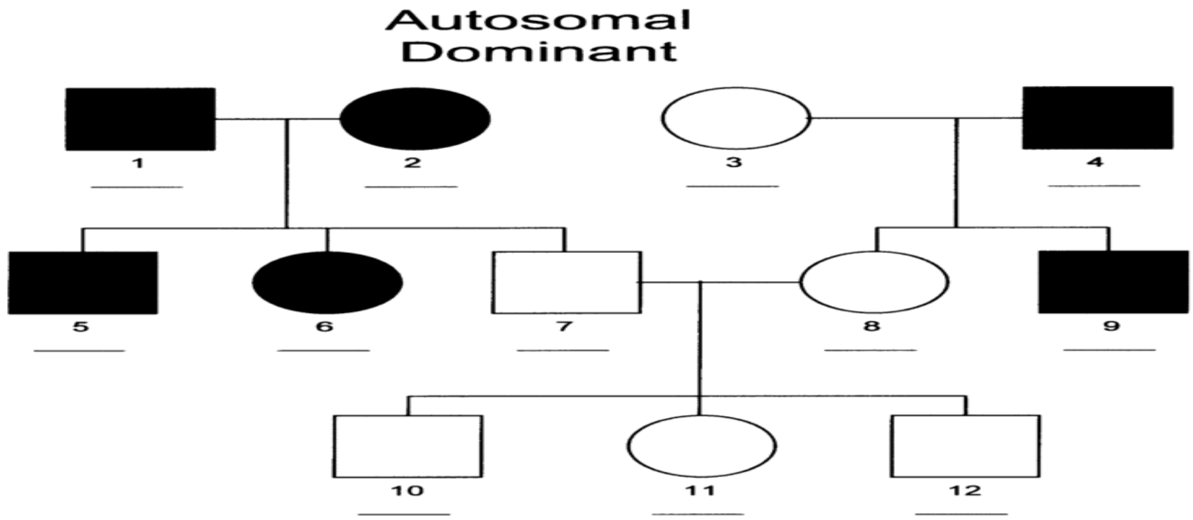
3. Risk Factors

Non-modifiable

- HTT gene mutation
- Family history
- Number of CAG repeats (greater repeats often mean earlier onset)

Modifiable

No confirmed lifestyle factors are known to prevent Huntington's disease because it is a genetic disorder, although exercise, good nutrition, and supportive care may improve quality of life.



4. Brain Regions Affected

Caudate Nucleus

One of the earliest and most severely affected regions.

Functions:

- Movement control
- Learning
- Cognition

Putamen

Responsible for:

- Motor coordination
- Habit learning

Globus Pallidus

Regulates voluntary movement.

Cerebral Cortex

Progressive degeneration causes:

- Cognitive decline
- Executive dysfunction

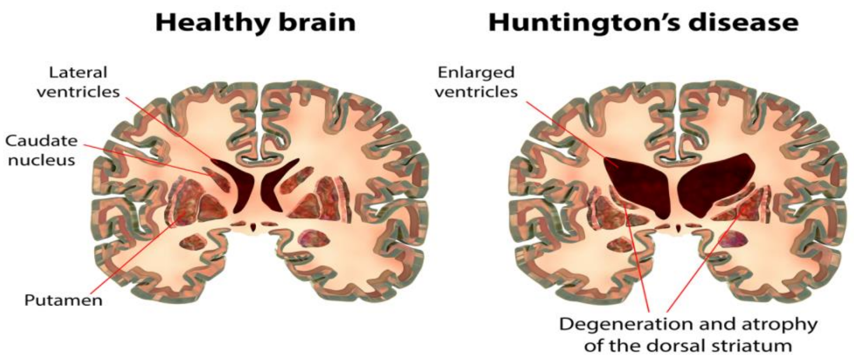
Thalamus

Motor information relay becomes disrupted.

Later Disease Stages

Degeneration spreads throughout:

- Basal ganglia
- Cerebral cortex
- White matter



5. Anatomy: Healthy vs Huntington's Brain

Discuss:

Healthy Brain

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Early Huntington's Disease

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Advanced Huntington's Disease

Key changes:

- Atrophy of the caudate nucleus
- Atrophy of the putamen
- Enlarged lateral ventricles
- Cortical thinning

6. Pathophysiology

HTT Gene Mutation

Normal HTT Gene

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CAG repeat expansion

↓

Mutant huntingtin protein

↓

Protein misfolding

↓

Protein aggregation

↓

Neuronal dysfunction

↓

Cell death

Protein Aggregation

Mutant huntingtin accumulates inside neurons, disrupting normal cellular function.

Excitotoxicity

Excess glutamate activity

↓

Calcium overload

↓

Neuronal injury

↓

Cell death

Mitochondrial Dysfunction

- Reduced ATP production
- Increased oxidative stress
- Energy failure

Neuroinflammation

Activation of:

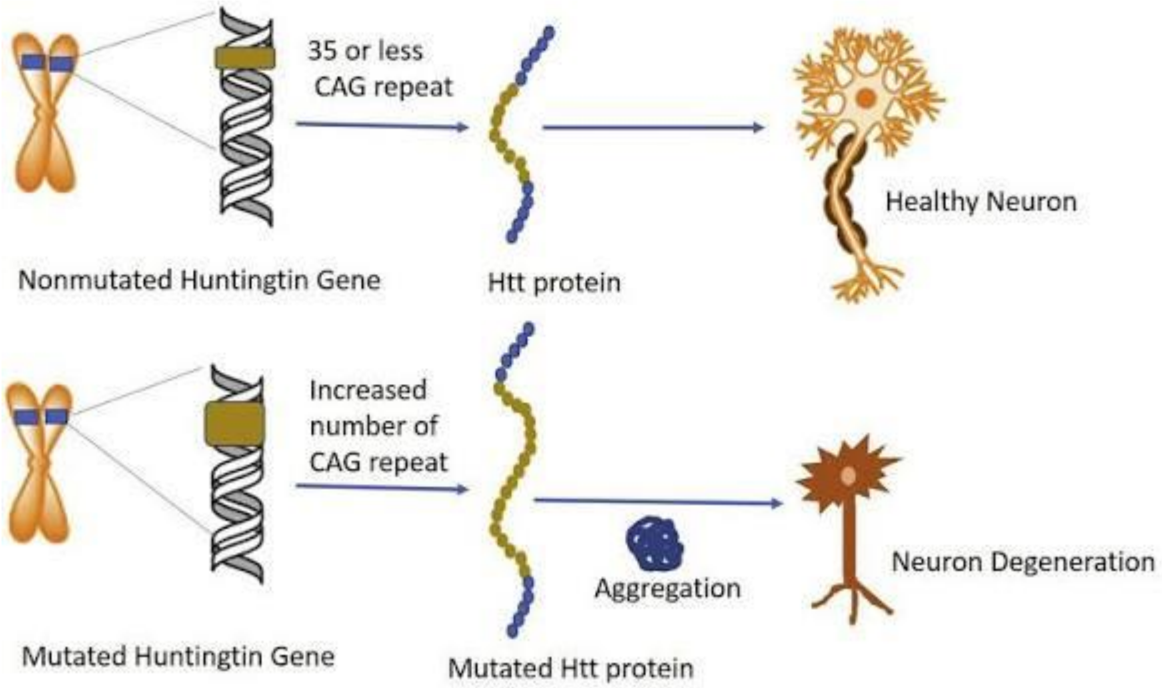
- Microglia
- Astrocytes

Release of inflammatory mediators contributing to neuronal degeneration.

Neuronal Loss

Most severe in:

- Medium spiny neurons
- Caudate nucleus
- Putamen



7. Genetics

HTT Gene

Located on chromosome 4.

Encodes the huntingtin protein.

CAG Repeat Expansion

Normal:

- 10–26 repeats

Intermediate:

- 27–35 repeats

Reduced penetrance:

- 36–39 repeats

Disease-causing:

- 40 or more repeats

Greater repeat numbers are often associated with earlier disease onset (**anticipation**).

Inheritance

- Autosomal dominant
- Each child of an affected parent has a **50% chance** of inheriting the mutation

8. Symptoms

Motor Symptoms

- Chorea (involuntary dance-like movements)
- Dystonia
- Poor coordination
- Balance problems
- Gait disturbances
- Difficulty swallowing (dysphagia)
- Difficulty speaking (dysarthria)

Cognitive Symptoms

- Memory impairment
- Reduced attention
- Poor judgment
- Executive dysfunction
- Dementia (later stages)

Psychiatric Symptoms

- Depression
- Anxiety
- Irritability
- Obsessive behaviors
- Aggression
- Apathy
- Psychosis (less common)

9. Clinical Stages

Preclinical Stage

Positive genetic test

No obvious symptoms

Early Stage

- Mild chorea
- Mood changes
- Difficulty concentrating

Middle Stage

- Significant motor impairment
- Cognitive decline
- Increasing dependence

Advanced Stage

- Severe motor disability
- Inability to walk
- Severe dementia
- Complete dependence

10. Diagnosis

Clinical Assessment

- Family history
- Neurological examination
- Psychiatric assessment
- Cognitive evaluation

Genetic Testing

Detection of CAG repeat expansion in the HTT gene.

This is the definitive diagnostic test.

Cognitive Testing

- Mini-Mental State Examination (MMSE)
- Montreal Cognitive Assessment (MoCA)

11. Neuroimaging

MRI

Findings:

- Caudate atrophy
- Putamen atrophy
- Enlarged ventricles
- Cortical atrophy

CT Scan

May demonstrate caudate atrophy in later disease stages.

PET Scan

Reduced glucose metabolism in the basal ganglia and cortex.

12. Biomarkers

Current research focuses on:

- Mutant huntingtin protein in CSF
- Neurofilament Light Chain (NfL)
- Plasma biomarkers
- MRI volumetric measurements

No routine clinical biomarker is currently used for monitoring disease progression.

13. Treatment

Medications for Chorea

- Tetrabenazine
- Deutetrabenazine

Antipsychotics

- Risperidone
- Olanzapine

Antidepressants

Selective serotonin reuptake inhibitors (SSRIs) may help manage depression and anxiety.

Supportive Therapy

- Physiotherapy
- Occupational therapy
- Speech therapy
- Nutritional support
- Psychological counseling

14. Latest Research

- Huntingtin-lowering therapies (antisense oligonucleotides)
- Gene-editing approaches (CRISPR)
- RNA interference (RNAi)
- Stem cell therapy
- Neuroprotective drugs
- AI-assisted disease progression prediction
- Biomarker development using blood and CSF

15. Future Directions

- Gene silencing therapies
- CRISPR-based gene editing
- RNA-targeted therapeutics
- Stem cell transplantation
- Personalized medicine
- Digital biomarkers
- AI for early disease prediction

16. References

- PubMed
- Nature Reviews Neurology
- Nature Neuroscience
- Neuron
- The Lancet Neurology
- Brain
- Journal of Huntington's Disease
- National Institute of Neurological Disorders and Stroke (NINDS)
- Huntington's Disease Society of America (HDSA)
- World Health Organization (WHO)